AUTOMATED NETWORK REQUEST MANAGEMENT IN SERVICENOW

Complete Project Documentation

ServiceNow implementation using Service Portal, Service Catalog, Flow Designer, approvals, custom tables, access controls and notifications.

DOCUMENT PROFILE	DETAIL
Project	Automated Network Request Management
Platform	ServiceNow
Primary interface	Service Portal
Core workflow	Catalog submission → Request records → Approval → Network Task → Status update → Notification
Document type	Functional and technical project documentation

Prepared as a consolidated description of the configured solution and the evidence available in the project screenshots.

Document Control & Contents

Item	Description
Project title	Automated Network Request Management in ServiceNow
Scope	Request intake, data capture, approval, task fulfilment, status tracking and notifications
Evidence basis	Configuration screenshots and outputs included in the supplied 27-page source document
Terminology	REQ = Request; RITM = Requested Item; ACL = Access Control List

Contents

- 1. Executive Summary
- 2. Project Overview and Objectives
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- 7. Phase 3 – Service Portal and UI Configuration
- 8. Phase 4 – Automation, Security and Testing
- 9. Phase 5 – Deployment and Handover
- 10. End-to-End Functional Output
- 11. Conclusion, Limitations and Future Enhancements

Note: The supplied source describes a configured ServiceNow solution and includes screenshots of records, approvals, flows, portal forms and notifications. This revision improves structure, consistency and technical wording without claiming configurations that are not evidenced.

1. Executive Summary

The Automated Network Request Management project provides a structured way for users to submit network-related service requests through a ServiceNow Service Portal. The solution captures request information using a Service Catalog item and catalog variables, creates the corresponding ServiceNow request records, routes the request through an approval process, and creates a Network Task for fulfilment after approval.

Flow Designer coordinates the configured processing steps, including reading catalog variables, creating a Network Database record, requesting approval, handling approval outcomes, creating or updating Network Tasks, updating status information and triggering email notifications.

The supplied evidence includes custom-table records, approval records, a Service Portal request form, Flow Designer configurations, ACL configurations, a successful submission output and email notification examples.

Key outcomes

- A centralized Service Portal entry point for network requests.
- Structured collection of requester, device and network details.
- A configured approval step before fulfilment.
- Automated Network Task creation for the fulfilment team after approval.
- Status tracking and email notifications at configured workflow stages.
- Access restrictions applied through ACL configurations.

Implementation boundary: The source document uses the phrase “Data Migration” in the Phase 4 heading, but it does not describe a migration source, mapping, migration execution or reconciliation. This revised document therefore labels the phase “Automation, Security and Testing” and does not claim that data migration was performed.

2. Project Overview and Objectives

2.1 Problem statement

Network-related service requests need a consistent intake and tracking process. A manual or fragmented process can make it difficult to capture complete information, route requests for approval, assign fulfilment work and communicate progress. This project addresses that need by configuring a ServiceNow-based request lifecycle.

2.2 Project objectives

- Provide a single catalog item for submitting network requests.
- Collect required information through structured catalog variables.
- Validate mandatory fields and populate the Requested For field with the current user.
- Create and relate the required request, database, approval and task records.
- Automate approval routing and fulfilment task creation using Flow Designer.
- Keep requesters and relevant users informed through configured notifications.
- Protect custom data using role- and rule-based ACL configurations.

2.3 In-scope capabilities

Capability	Purpose
Service Catalog item	Provides the Network Request submission form.
Catalog variables and variable set	Capture and organize request details.
UI Policy and Catalog Client Script	Control client-side form behaviour.
Custom tables	Store Network Database and Network Task information.
Flow Designer and approvals	Coordinate approval and fulfilment processing.
ACLs and notifications	Control data access and communicate workflow updates.

3. Requirements and Stakeholders

3.1 Stakeholders and responsibilities

Stakeholder	Responsibility
Requester / End User	Submits a Network Request and supplies accurate details.
Manager / Approver	Reviews the request and records an approval decision.
Network Fulfilment Team	Works on the generated Network Task and updates its progress.
ServiceNow Administrator	Configures catalog items, variables, tables, flows, ACLs and notifications.

3.2 Functional requirements

1. The user can open the Network Request catalog item through the Service Portal.
2. The form captures the required requester, contact, device, connection and address information.
3. Mandatory fields are validated before submission.
4. Requested For is populated automatically with the current user.
5. ServiceNow generates the Request (REQ) and Requested Item (RITM) records.
6. The configured flow creates a corresponding Network Database record.
7. An approval request is generated and the configured approver can approve or reject it.
8. An approved request results in a Network Task for fulfilment.
9. The workflow updates relevant status information and sends configured email notifications.

3.3 Non-functional considerations

- Usability: present related fields together and make required inputs clear.
- Security: restrict read, create, write and delete operations according to configured ACL rules.
- Data integrity: preserve relationships between request, approval and task records.
- Maintainability: keep configuration components organized and document navigation paths.
- Traceability: use request numbers and related records to follow processing.

4. Solution Architecture and Workflow

The solution uses the Service Portal as the user-facing entry point. Service Catalog captures the request, standard ServiceNow request records represent the submission, a custom Network Database table stores the structured request details, and Flow Designer coordinates approval and fulfilment. The Network Task table represents work assigned to the Network Team.

4.1 Logical workflow

Stage	Processing activity	Expected record / result
1. Submit	User completes and submits the catalog form.	REQ and RITM are generated.
2. Capture	Flow reads the submitted catalog variables.	Network Database record is created.
3. Approve	Configured approver reviews the request.	Approval state is recorded.
4. Fulfil	Approved request creates a Network Task.	Task is available to the fulfilment team.
5. Update	Task and request progress are updated.	Current status is reflected in records.
6. Notify	Configured notifications are triggered.	Relevant users receive email updates.

4.2 End-to-end workflow

Service Portal → Network Request → REQ / RITM → Network Database → Approval → Network Task → Task Completion → Status Update → Email Notification

The rejection path is also described in the source as a configured approval branch. The precise rejection-state labels and any rejection notification content should be confirmed against the live instance before operational handover.

5. Phase 1 – Requirement Analysis and Planning

5.1 Requirement discovery

- Identify the need for a structured network service request process.
- Define the information users must provide at submission time.
- Identify the requester, approver, fulfilment team and administrator roles.
- Define the lifecycle from submission through validation, approval, task fulfilment, status update and notification.

5.2 Planned ServiceNow components

- Service Catalog and Network Request catalog item
- Catalog variables and variable set
- Network Database and Network Task custom tables
- Flow Designer and approval records
- Catalog UI Policy and Catalog Client Script
- Business Rules, notifications and ACLs
- Update Sets for configuration packaging

5.3 Implementation roadmap

Phase	Focus	Main deliverable
1	Requirement analysis and planning	Scope, stakeholders and roadmap
2	Backend development and configuration	Custom tables, records and approval relationship
3	UI/UX development and customization	Catalog item, variables and portal form
4	Automation, security and testing	Flow, ACLs and end-to-end validation
5	Deployment, documentation and presentation	Troubleshooting, handover and demonstration

6. Phase 2 – Backend Development and Configuration

6.1 Data architecture

Two custom data structures are described. The Network Database table stores structured request information. The Network Task table represents the operational work generated for the Network Team. The records are used alongside standard ServiceNow request and approval records.

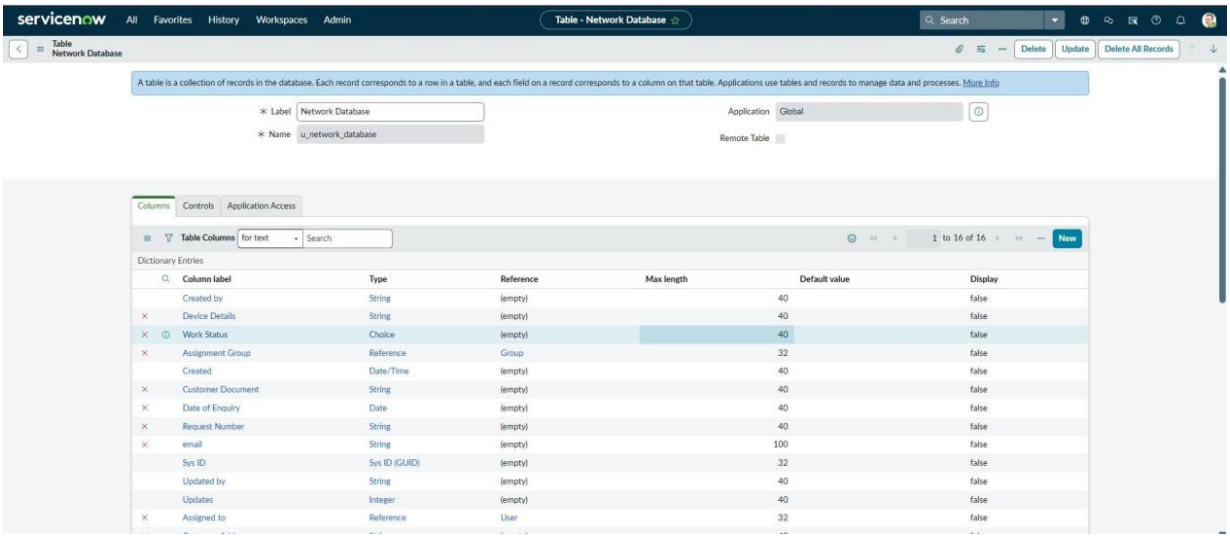


Figure 6.1. Network Database table/list configuration (source evidence, page 9).

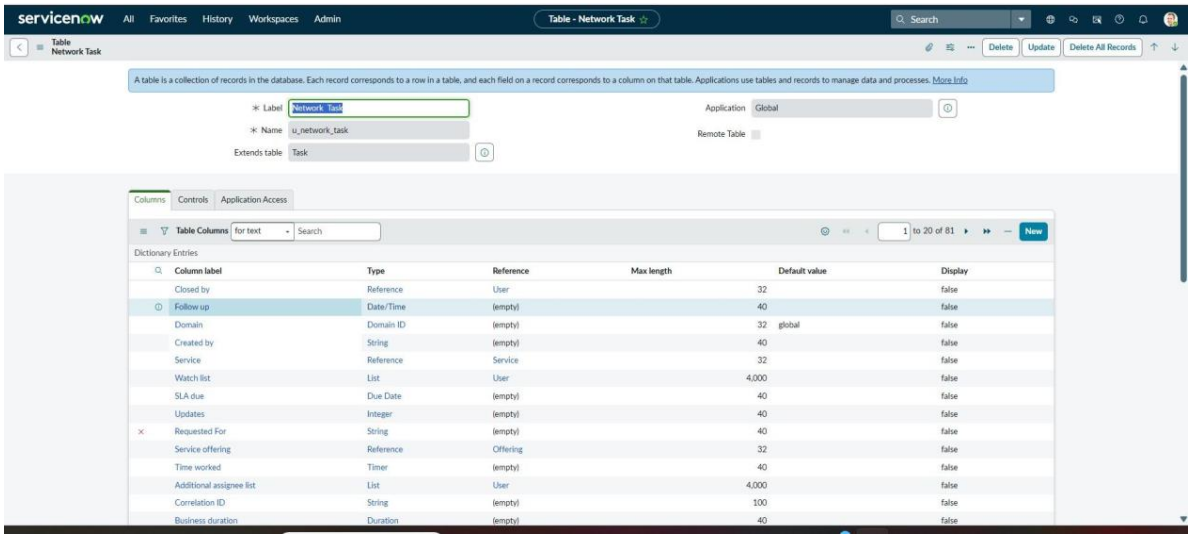


Figure 6.2. Network Task records and related information (source evidence, page 9).

6.2 Network Task

A Network Task is generated as part of the configured request lifecycle after approval. It carries operational information needed by the Network Team and supports subsequent task-status updates.

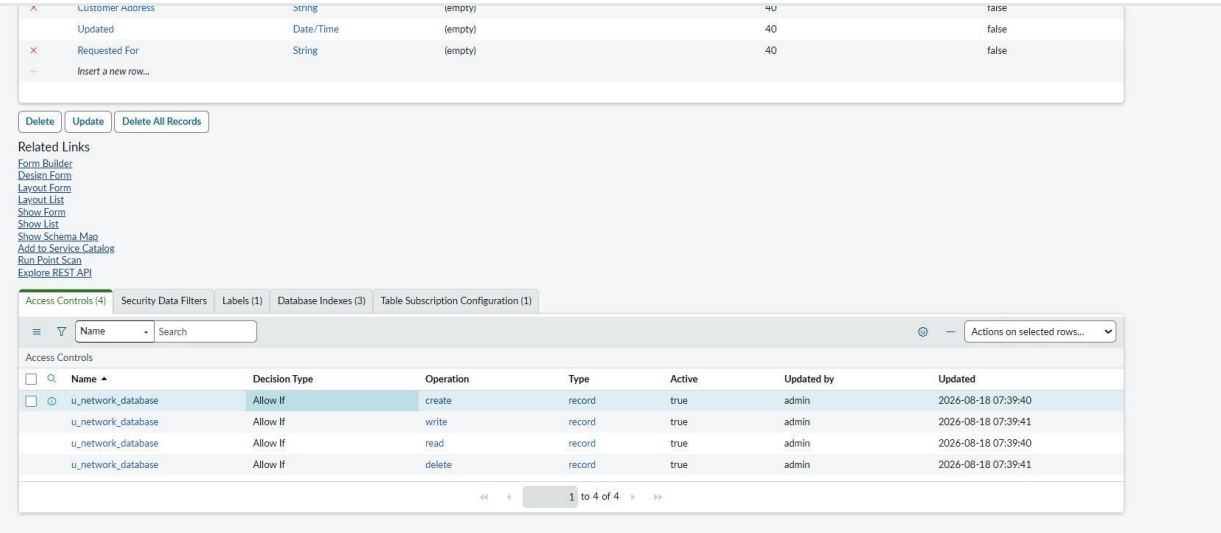


Figure 6.3. Network Task record/output (source evidence, page 10).

6.3 Approval relationship and output

The source documents an Approval Request relationship associated with the Network Database record. The related list provides visibility into approval records connected with the request. The supplied approval output shows records with an approver and an Approved state.

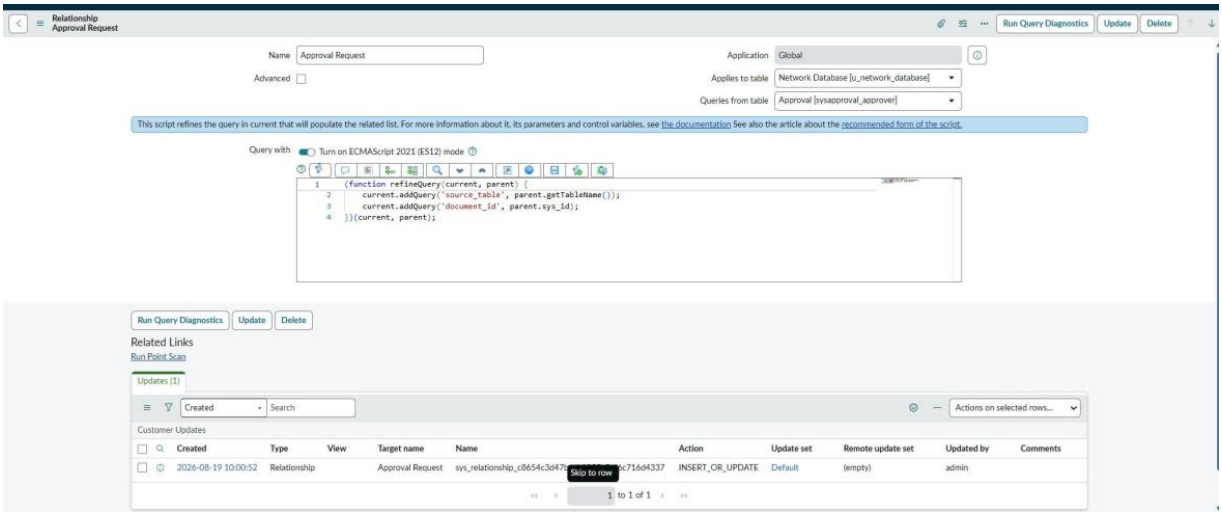


Figure 6.4. Approval Request relationship configuration (source evidence, page 10).

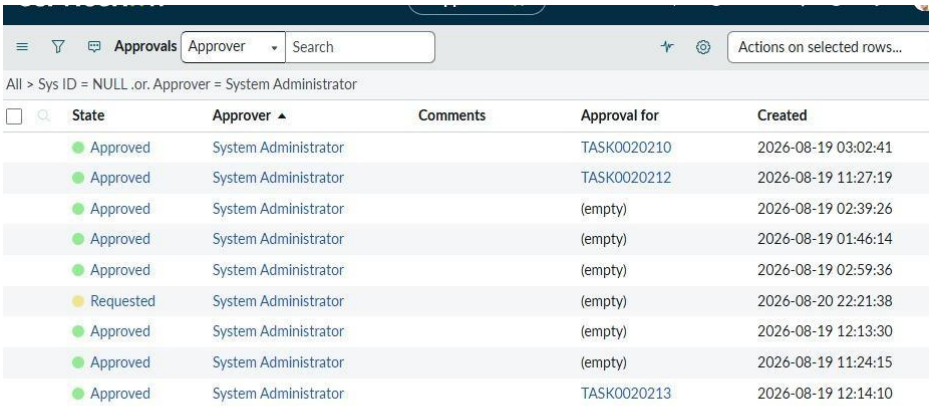


Figure 6.5. Approval records showing state and approver (source evidence, page 11).

7. Phase 3 – Service Portal and UI Configuration

7.1 Network Request catalog item

The Network Request catalog item is created under Service Catalog → Catalog Definitions → Maintain Items. The item acts as the primary submission form for users.

1. Open Maintain Items and select New.
2. Create the Network Request catalog item.
3. Enter the required item details and save the record.
4. Open the item and configure its variables.

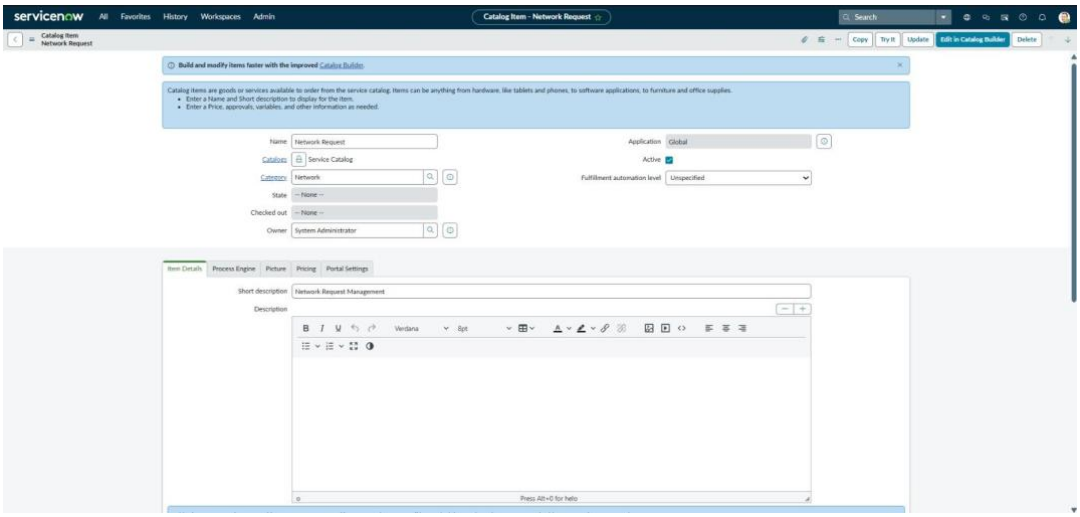


Figure 7.1. Catalog item creation/configuration evidence (source evidence, page 12).

7.2 Variables and variable set

Catalog variables collect the request information. The source identifies ten variables and one variable set. Fields are configured with suitable variable types, mandatory settings and display order. Choice values include device options such as Laptop, Mobiles and Others.

1. Open the Network Request catalog item and navigate to Variables.
2. Create the required variables with appropriate types.
3. Configure mandatory fields, display order and choice values.
4. Create a Variable Set, add related variables and associate it with the catalog item.

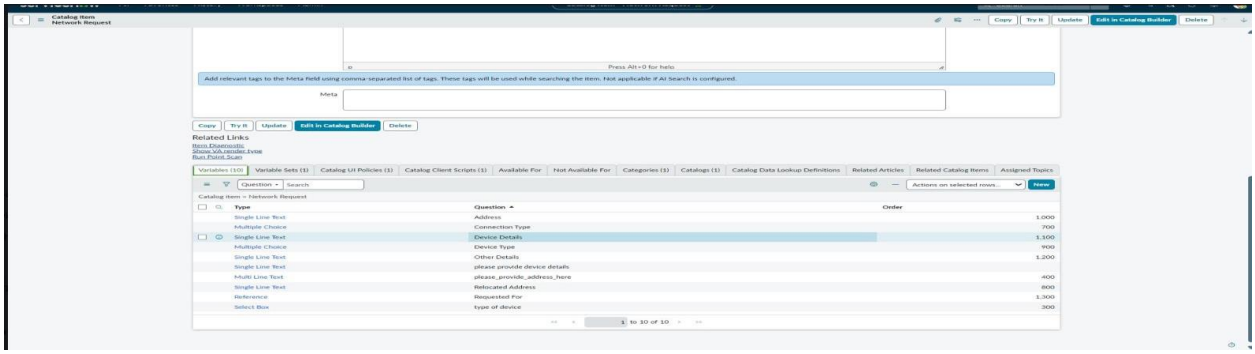


Figure 7.2. Variable configuration evidence (source evidence, page 13).

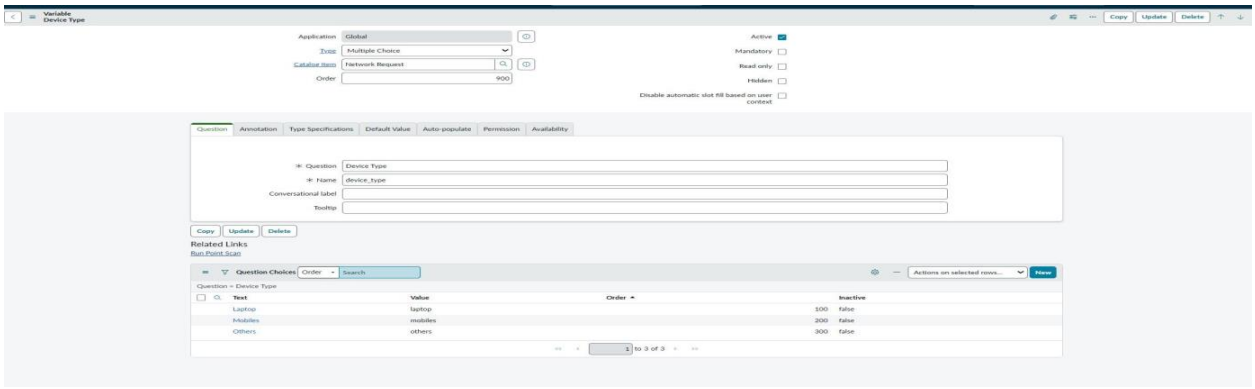


Figure 7.3. Variable creation evidence (source evidence, page 13).

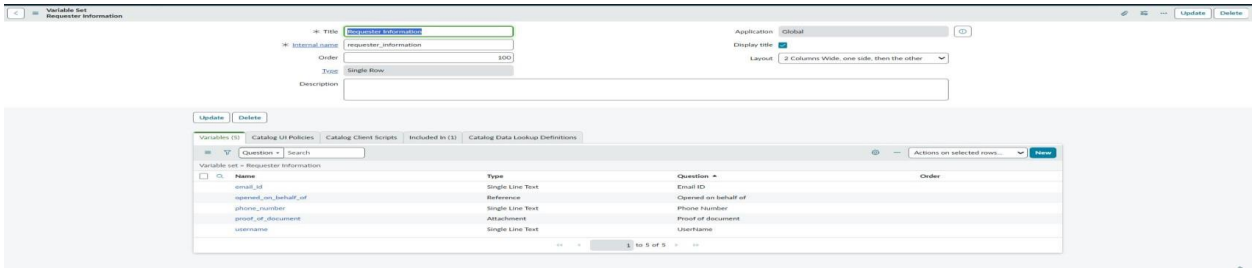


Figure 7.4. Variable Set configuration evidence (source evidence, page 14).

7.3 Navigation and usability

- Log in to the ServiceNow portal.
- Search for and open Network Requests.
- Open the Network Request catalog item.
- Enter the required information and submit the request.
- Use the generated request number to track progress.

The supplied portal screenshots show grouped requester and device information, mandatory-field indicators, submission controls, Requested For information and attachment support.

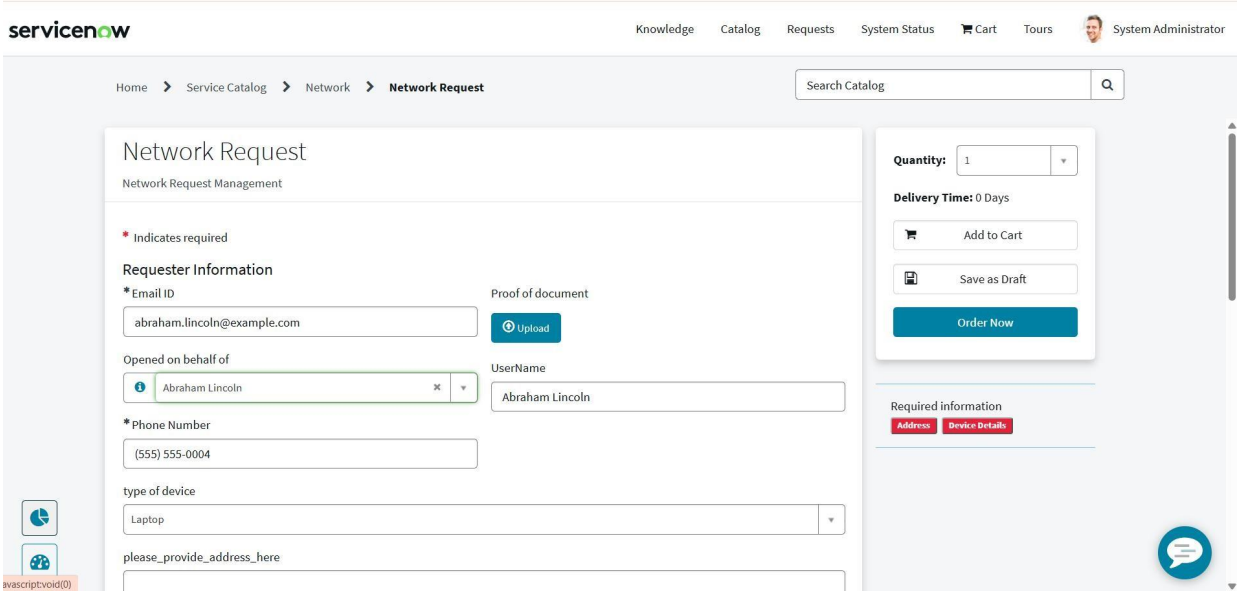


Figure 7.5. Network Request form and usability evidence (source evidence, page 15).

7.4 UI logic and client-side configuration

The catalog item is documented with ten variables, one variable set, one Catalog UI Policy and one Catalog Client Script. The UI Policy is named “Types of devices is others”. It is described as controlling form behaviour according to the selected device type, running on load and reversing its action when the condition becomes false.

Component	Documented purpose
Catalog UI Policy	Controls field visibility or behaviour based on the selected device type.
Catalog Client Script	Performs client-side actions when the catalog form loads or the user interacts with it.
Catalog availability	Controls where and to whom the catalog item is available.

Navigation reference: Service Catalog → Catalog Definitions → Maintain Items → Network Request. Open the item and access the relevant UI Policy or Catalog Client Script related list.

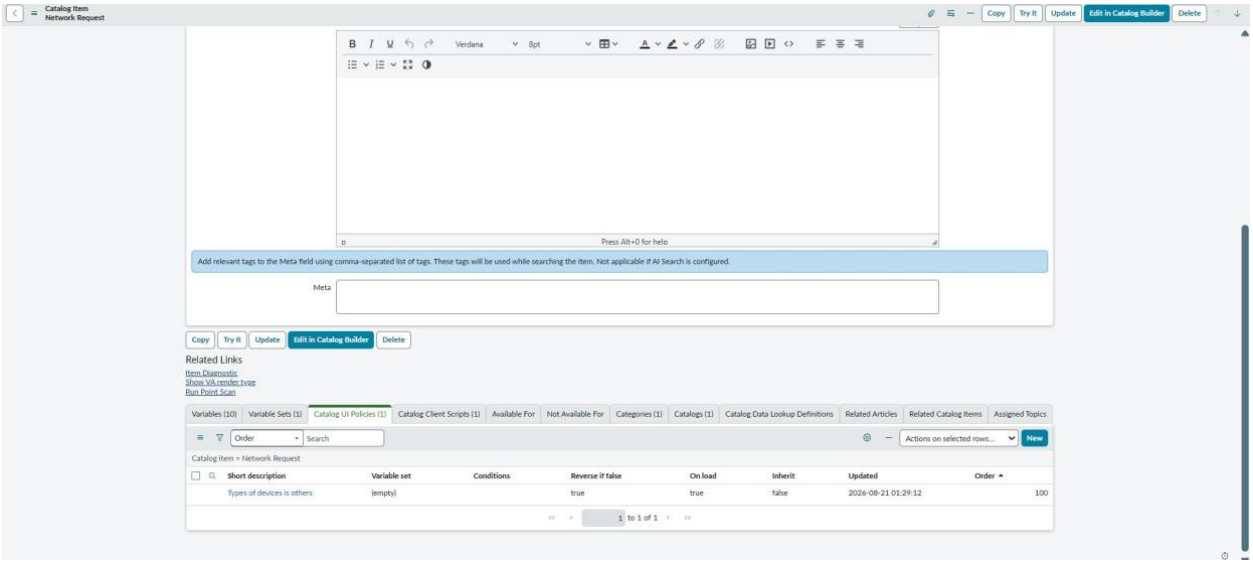


Figure 7.6. Catalog item variables, UI Policy and Client Script configuration (source evidence, page 17).

8. Phase 4 – Automation, Security and Testing

8.1 Flow Designer automation

Flow Designer coordinates the configured request lifecycle. The source describes a flow that reads catalog variables, creates the Network Database record, sends an approval request, handles approved and rejected branches, creates and updates Network Tasks, updates Network Database status and sends email notifications.

1. Open ServiceNow → All → Flow Designer.
2. Open the configured Network Request flow.
3. Verify the trigger and workflow actions.
4. Review approval and fulfilment actions, including the approval branches.
5. Confirm that a Network Task is created after approval.
6. Save and activate the flow, then validate its execution.

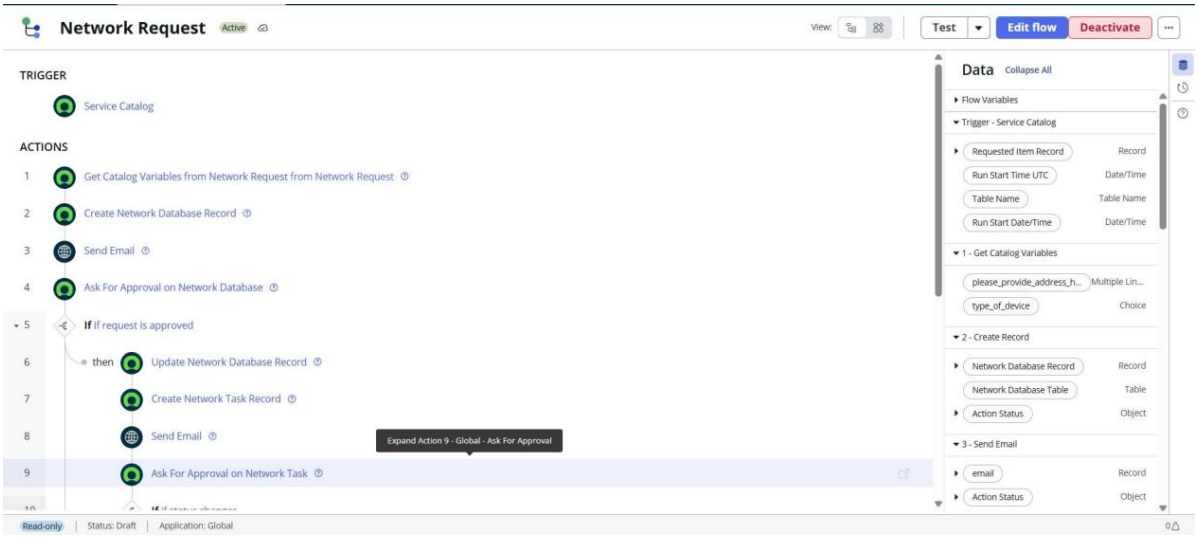


Figure 8.1. Network Request Flow Designer automation (source evidence, page 18).

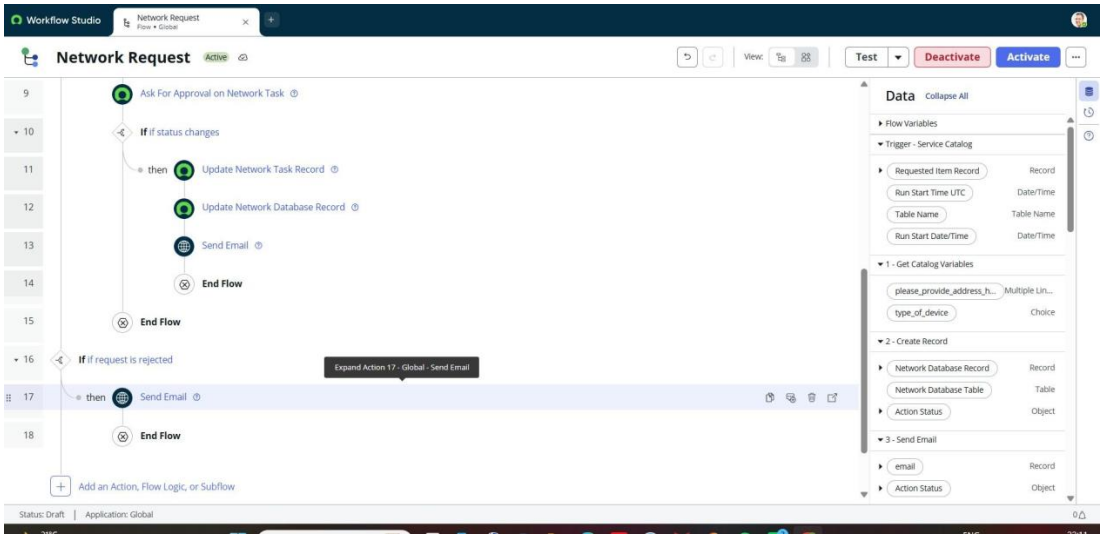


Figure 8.2. Flow execution and approval/update branches (source evidence, page 18).

8.2 Access control

Access control is implemented through ACL configurations. Read, create, write and delete operations are controlled according to the configured rules. The source describes checking permissions for Network Request data and ensuring that authorized users, approvers and fulfilment users have the access required for their responsibilities.

- Navigate to All → System Security → Access Control (ACL).
- Review the ACLs associated with the custom Network Request data.
- Verify authorized create and update access.
- Verify the access needed by approvers and fulfilment users.
- Test both permitted and restricted access using appropriate test accounts.

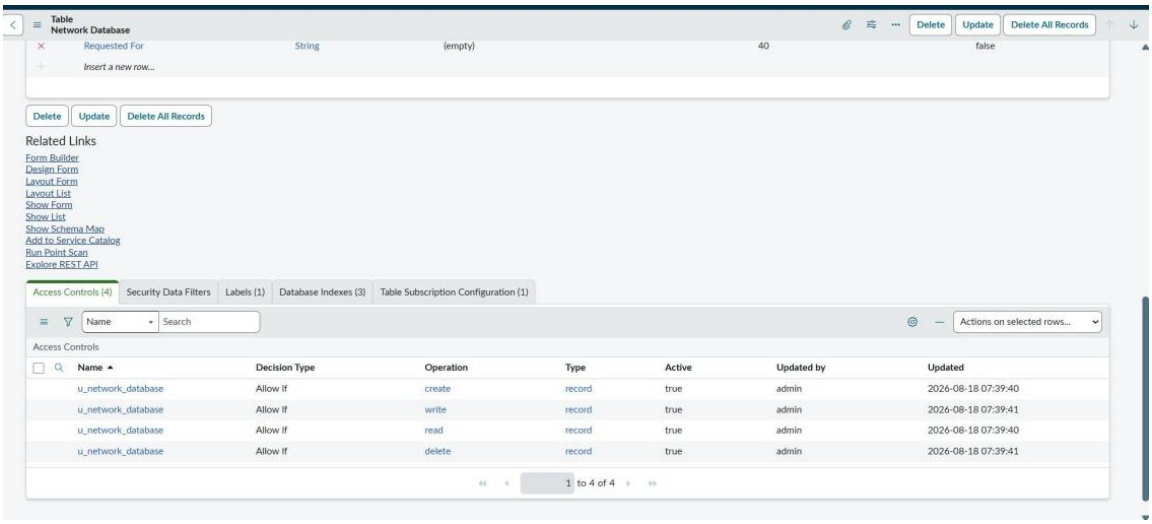


Figure 8.3. Access Control configuration evidence (source evidence, page 19).

8.3 Quality assurance and end-to-end testing

The source describes end-to-end testing of request submission, record creation, approval processing, task creation, status updates and email notification.

Test ID	Test scenario	Expected result
TC-01	Submit a valid Network Request.	REQ and RITM are generated.
TC-02	Check submitted variable values.	Values are stored in the corresponding Network Database record.
TC-03	Submit a request that requires approval.	Approval record is generated for the configured approver.
TC-04	Approve the request.	Network Task is created for fulfilment.
TC-05	Update/complete the Network Task.	Task and request status reflect the configured progression.
TC-06	Check configured notifications.	Relevant email notification is generated.

The table formalizes test scenarios described in the source. Actual pass/fail results should be recorded during execution; the screenshots provide supporting evidence for selected successful outputs.

8.4 Testing Evidence and Data Integrity

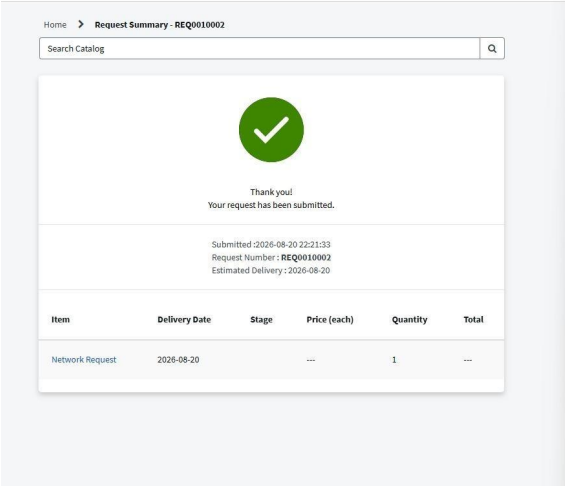


Figure 8.4. Successful Network Request submission and generated request number (source evidence, page 20).

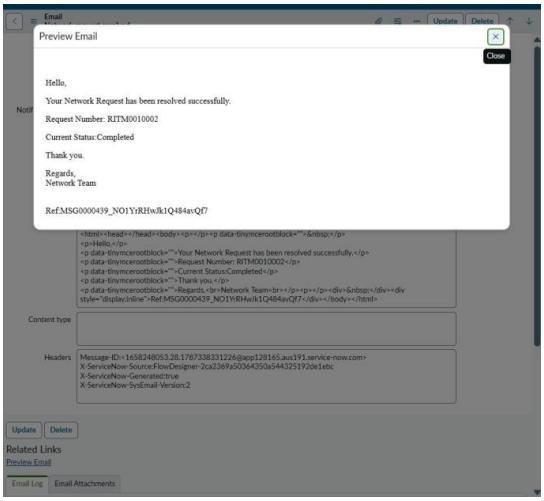


Figure 8.5. Email notification output (source evidence, page 20).

Data integrity checks include confirming that submitted values are stored correctly, that request, approval and task records are related correctly, and that required information is not missing or incorrectly mapped.

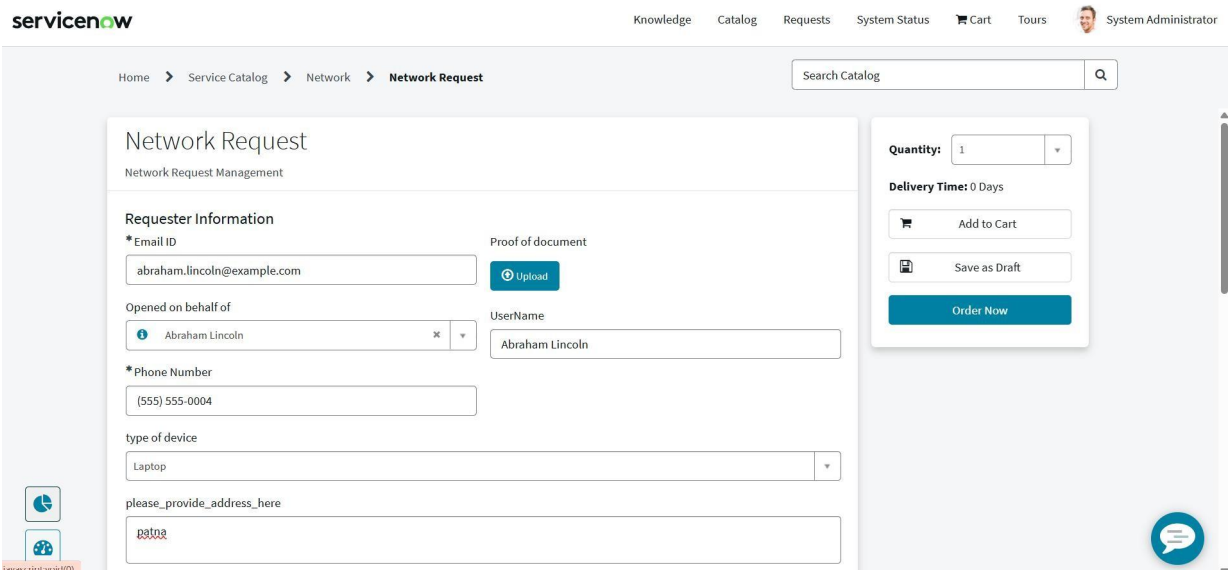


Figure 8.6. Required fields and requester information (source evidence, page 21).

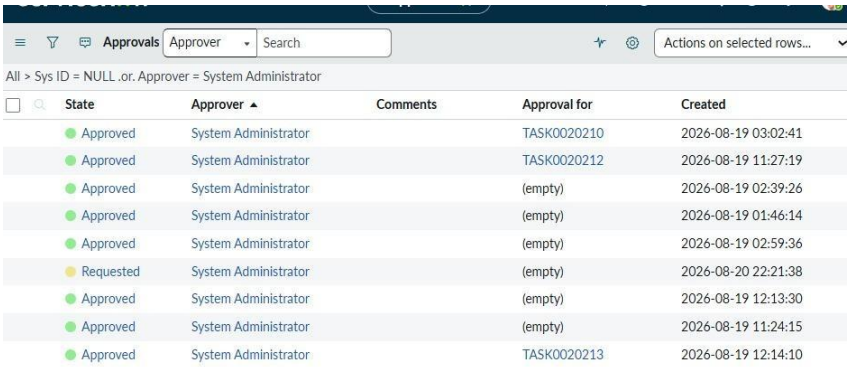


Figure 8.7. Approval-state evidence used for validation (source evidence, page 22).

9. Phase 5 – Deployment, Documentation and Handover

The source describes Phase 5 as the final documentation and presentation stage. The activities below are presented as a handover checklist. They should not be interpreted as proof of a production deployment unless deployment records are separately available.

9.1 Final verification

1. Review the complete Network Request workflow.
2. Check the catalog item, variables, approvals, tasks and notifications for configuration errors.
3. Verify that the configured flow executes correctly.
4. Resolve configuration or validation issues.
5. Perform a final end-to-end verification.

9.2 Documentation and handover

- Prepare a functional overview describing the purpose and request lifecycle.
- Document the technical components, custom tables and relationships.
- Record navigation paths and configuration steps for administrators.
- Include relevant screenshots as configuration and output evidence.
- Prepare a demonstration showing request submission, generated records, approval, task creation and final status.

9.3 Demonstration sequence

1. Open the Network Request catalog item in the Service Portal.
2. Enter request details and submit using Order Now.
3. Show the generated REQ/RITM and Network Database record.
4. Show the approval record and decision.
5. Show the Network Task created for fulfilment.
6. Show the final task/request status and notification output.

9.4 Scalability and future plan

- Extend the catalog with additional network service request types.
- Introduce additional approval levels where business rules require them.
- Automate more fulfilment activities through Flow Designer.
- Improve reporting and monitoring for request volumes, ageing and fulfilment performance.

10. End-to-End Functional Output

The implemented flow is documented as follows:

1. The user opens Network Request through the Service Portal.
2. The requester enters the required network and device details.
3. The user submits the form using Order Now; ServiceNow generates a Request (REQ) and Requested Item (RITM).
4. The submitted details are recorded in the configured Network Database record.
5. An approval request is generated for the configured approver.
6. The approver reviews and approves or rejects the request.
7. When approved, a Network Task is created for the fulfilment team.
8. The Network Task and Network Database status are updated as processing continues.
9. After successful completion, the request is marked completed/resolved according to the configured process and a completion email is generated for the configured recipient.

Service Portal → Network Request → REQ/RITM → Network Database → Approval → Network Task → Task Completion → Status Update → Email Notification

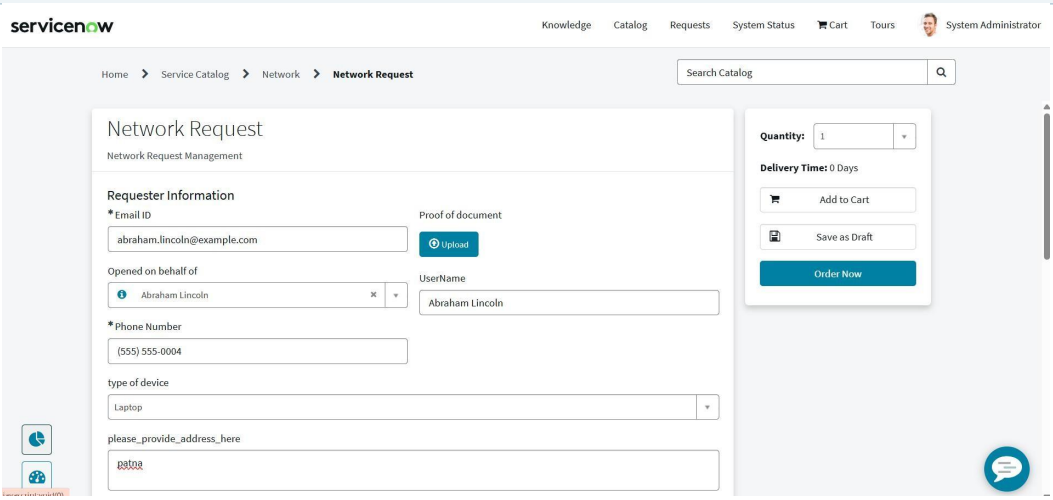


Figure 10.1. Request submission form/output evidence (source evidence, page 25).

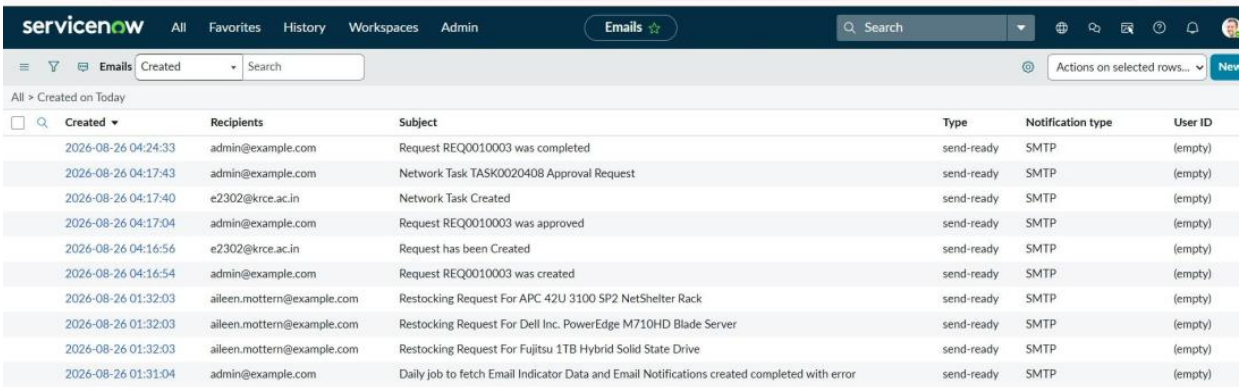


Figure 10.2. Email output evidence (source evidence, page 26).

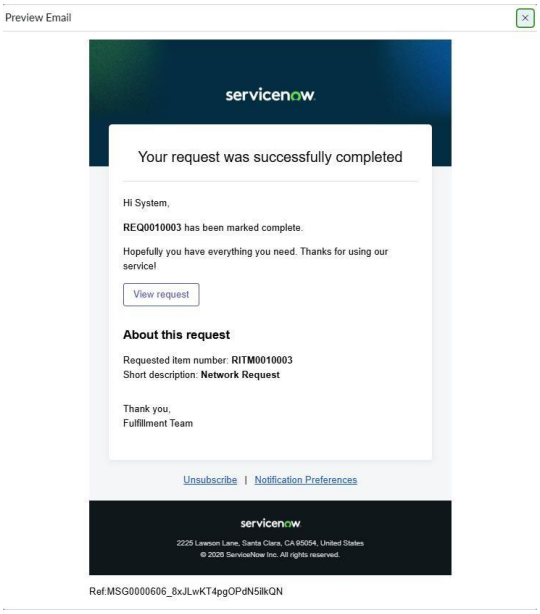


Figure 10.3. Completion notification email example (source evidence, page 26).

11. Conclusion

The Automated Network Request Management project demonstrates a structured ServiceNow process for handling network-related service requests. It brings together Service Catalog, catalog variables, custom tables, approval records, Flow Designer, Network Tasks, ACLs and email notifications.

The documented functional path covers request submission, request-record generation, approval, task fulfilment, status updates and requester notification. The screenshots provide evidence of the configured portal form, custom-table records, approval information, flow configuration, access-control setup and selected successful outputs.

Benefits

- Centralized request submission and consistent data capture.
- A defined approval stage before fulfilment.
- Automated task creation and improved work visibility.
- Traceability through request, approval and task records.
- Email communication at configured workflow stages.
- A foundation that can be extended to additional network services.

Items to confirm before formal production handover

- Confirm the exact ACL roles and table/field permissions in the target instance.
- Record the final test execution date, tester, actual result and any defects.
- Confirm rejection-path status handling and requester notification behaviour.
- Confirm deployment method, update-set contents and target-instance validation.
- Add owner/contact details, version number and approval/sign-off information if required by the organization.

End of Document